

Exercice Pratique AWS RDS Objectif : Créer une base de données MySQL sur RDS et la connecter à votre instance EC2 via Node.js Durée : 45-60 minutes

Configuration RDS

- Choisir MySQL comme moteur
- Version : MySQL 8.0.28
- Template : Free tier
- DB Instance Identifier : myapp-database
- Master username : admin
- Générer un mot de passe sécurisé
- Instance type : db.t3.micro

Choose a database creation method

☐ Standard create
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☒ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Configuration

Engine type [Info](#)

☐ Aurora (MySQL Compatible)



☐ Aurora (PostgreSQL Compatible)



☒ MySQL



☐ PostgreSQL



☐ MariaDB



☐ Oracle



☐ Microsoft SQL Server



Edition

☒ MySQL Community

DB instance size

1. Sur l'EC2, installer le client MySQL et créer la base de données

Installation du client MySQL

sudo dnf install mariadb105 -y

```
Installed:
  mariadb-connector-c-3.1.13-1.amzn2023.0.3.x86_64  mariadb-connector-c-config-3.1.13-1.amzn2023.0.3.noarch  mariadb105-3:10.5.23-1.amzn2023.0.1.x86_64  mariadb105-common-3:10.5.23-1.amzn2023.0.1.x86_64
  perl-DBD-MySQL-1.23-477.amzn2023.0.6.x86_64

Complete!
[ec2-user@ip-172-31-13-199 ~]#
```

Connexion à votre instance RDS

database-sabri.cr4yi8k40i7t.eu-west-3.rds.amazonaws.com

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations - new

Tags

Recommendations

Connectivity & security

Endpoint copied

database-sabrl-cr4yi8k40i7t.eu-west-3.rds.amazonaws.com

Port

3306

Networking

Availability Zone

eu-west-3a

VPC

vpw-099a97a6f9fdad592

Subnet group

rds-ec2-db-subnet-group-1

Subnets

subnet-0408444278d933b94

subnet-057478ed0ed18940

subnet-0aed4a427ab03d144

Network type

IPv4

Security

VPC security groups

rds-ec2-4 (sg-00b1bd41a9e614999)

Active

Publicly accessible

No

Certificate authority

rds-ca-rsa2048-g1

Info

Certificate authority date

May 26, 2061, 01:18 (UTC+02:00)

DB instance certificate expiration date

March 05, 2026, 14:52 (UTC+01:00)

```
mysql -h [VOTRE-ENDPOINT-RDS] -P 3306 -u admin -p
```

```
mysql -h database-sabri.cr4yi8k40i7t.eu-west-3.rds.amazonaws.com -P 3306 -u admin  
-p
```

```
#
~\##### Amazon Linux 2023
~~\#####\
~~\###|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~' '->
~~~~
~-.-
~/m/' -/-
```

Last login: Wed Mar 5 14:18:31 2025 from 35.180.112.83
ec2-user@ip-172-31-13-199 ~]\$ mysql -h database-sabri.cr4yi8k4oi7t.eu-west-3.rds.amazonaws.com -P 3306 -u admin -p
ERROR 1045 (28000): Access denied for user 'admin'@'172.31.13.199' (using password: NO)
ec2-user@ip-172-31-13-199 ~]\$ mysql -h database-sabri.cr4yi8k4oi7t.eu-west-3.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MySQL connection id is 42
Server version: 8.0.40 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>

Ouverture du port 3306

Edit inbound rules [info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type info	Protocol info	Port range info	Source info	Description - optional info	
sgr-007c80252138f7da	SSH	TCP	22	Custom	Q	0.0.0.0/0 X Delete
sgr-05c84101560c82b7b	MySQL/Aurora	TCP	3306	Custom	Q	0.0.0.0/0 X Delete
sgr-0fcb1acc8dd8f93de	HTTPS	TCP	443	Custom	Q	0.0.0.0/0 X Delete
sgr-0ba6db1a80da6b9ee	HTTP	TCP	80	Custom	Q	0.0.0.0/0 X Delete

[Add rule](#)

⚠ Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#) [Preview changes](#) [Save rules](#)

Une fois connecté à MySQL, créer la base et la table

CREATE DATABASE webappdb; USE webappdb;

```
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE webappdb; USE webappdb;
Query OK, 1 row affected (0.021 sec)

Database changed
MySQL [webappdb]> 
```

Création de la table users

CREATE TABLE users (id INT AUTO_INCREMENT PRIMARY KEY, username
VARCHAR(50) NOT NULL, email VARCHAR(100) NOT NULL, created_at TIMESTAMP
DEFAULT CURRENT_TIMESTAMP);

```
Database changed
MySQL [webappdb]> CREATE TABLE users ( id INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(50) NOT NULL, email VARCHAR(100) NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP );
Query OK, 0 rows affected (0.056 sec)

MySQL [webappdb]> 
```

Insertion de données de test

INSERT INTO users (username, email) VALUES ('john_doe', 'john@example.com'),
('jane_doe', 'jane@example.com'), ('bob_smith', 'bob@example.com');

```
Database changed
MySQL [webappdb]> CREATE TABLE users ( id INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(50) NOT NULL, email VARCHAR(100) NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP );
Query OK, 0 rows affected (0.056 sec)

MySQL [webappdb]> INSERT INTO users (username, email) VALUES ('john_doe', 'john@example.com'), ('jane_doe', 'jane@example.com'), ('bob_smith', 'bob@example.com');
Query OK, 3 rows affected (0.008 sec)
Records: 3 Duplicates: 0 Warnings: 0

MySQL [webappdb]> 
```

Vérifier que tout est bien créé

SELECT * FROM users;

```
MySQL [webappdb]> SELECT * FROM users;
+----+-----+-----+-----+
| id | username | email | created_at |
+----+-----+-----+-----+
| 1 | john_doe | john@example.com | 2025-03-05 14:38:47 |
| 2 | jane_doe | jane@example.com | 2025-03-05 14:38:47 |
| 3 | bob_smith | bob@example.com | 2025-03-05 14:38:47 |
+----+-----+-----+-----+
3 rows in set (0.001 sec)

MySQL [webappdb]> 
```

Quitter MySQL

Exit

```
MySQL [webappdb]> exit
Bye
[ec2-user@ip-172-31-13-199 ~]$ 
```

2. Sur votre instance EC2, installer Node.js

Installation de Node.js

curl -o- <https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.0/install.sh> | bash .

~/nvm/nvm.sh nvm install 16 nvm use 16

```
100 14984 100 14984 0 0 486k 0 --:--:-- --:--:-- --:--:-- 487k
=> Downloading nvm as script to '/home/ec2-user/.nvm'

=> Appending nvm source string to /home/ec2-user/.bashrc
=> Appending bash_completion source string to /home/ec2-user/.bashrc
=> Close and reopen your terminal to start using nvm or run the following to use it now:

export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh" # This loads nvm
[ -s "$NVM_DIR/bash_completion" ] && \. "$NVM_DIR/bash_completion" # This loads nvm bash_completion
[ec2-user@ip-172-31-13-199 ~]$ 
```

3. Créer un projet Node.js
mkdir myapp cd myapp npm init -y

```
Wrote to /home/ec2-user/myapp/package.json:

{
  "name": "myapp",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

[ec2-user@ip-172-31-13-199 myapp]$
```

Installation des dépendances

```
npm install express mysql2 dotenv
```

```
found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 8.19.4 -> 11.1.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.1.0
npm notice Run npm install -g npm@11.1.0 to update!
npm notice
[ec2-user@ip-172-31-13-199 myapp]$
```

Créer le fichier server.js

```
require('dotenv').config(); const express = require('express'); const mysql =
require('mysql2/promise'); const app = express(); const pool = mysql.createPool({ host:
votre-endpoint-rds, user: admin, password: votre-mot-de-passe, database: webappdb
}); app.get('/', async (req, res) => { try { const [rows] = await pool.query('SELECT * FROM
users'); res.send(`
```

```
[ec2-user@ip-172-31-13-199 myapp]$ require('dotenv').config(); const express = require('express'); const mysql = require('mysql2/promise'); const app = express(); const pool = mysql.createPool({ host: vot
-e: endpoint-rds, user: admin, password: votre-mot-de-passe, database: whappdb }); app.get('/', async (req, res) => { try { const [rows] = await pool.query('SELECT * FROM users'); res.send(
bash: syntax error near unexpected token `dotenv'
[ec2-user@ip-172-31-13-199 myapp]$
```

Liste des utilisateurs

```
rows.map(user => {
```

Utilisateur : \${user.username}

Email : \${user.email}

```
`.join(")} `"); } catch (error) { console.error(error); res.status(500).send('Erreur serveur');
}); app.listen(3000, () => { console.log('Serveur en cours d\'exécution sur le port 3000');
});
```

```
Utilisateur : ${user.username}
Email : ${user.email}

').join('') '); } catch (error) { console.error(error); res.status(500).send('Erreur serveur'); } )); app.listen(3000, () => { console.log('Serveur en cours d\'exécution sur le port 3000'); });
```

4. Lancer l'application

Lancer en arrière-plan

node server.js &

Noter le PID si vous devez arrêter le processus plus tard

Pour arrêter : kill

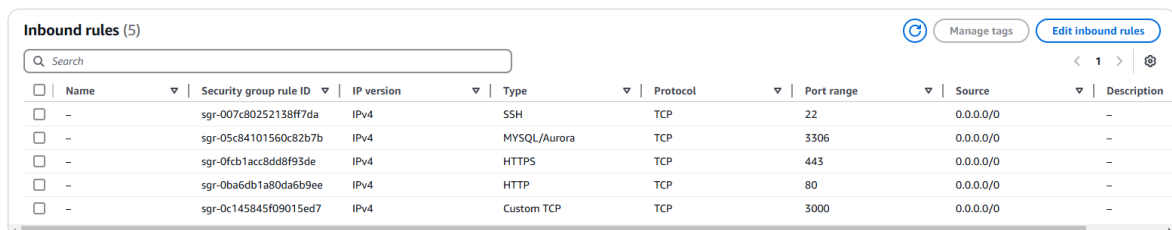
5. Vérifier que l'application fonctionne

Tester localement

curl <http://localhost:3000>

L'application sera accessible via

[http://\[IP-PUBLIQUE-EC2\]:3000](http://[IP-PUBLIQUE-EC2]:3000) N'oubliez pas : D'ouvrir le port 3000 dans le Security Group de l'EC2 De sauvegarder le PID quelque part si vous devez arrêter le processus Cette méthode est pour le développement/test uniquement, pas pour la production Pour arrêter l'application :



The screenshot shows the 'Inbound rules' section of an AWS Security Group. It contains a table with 5 rules. The last rule is a 'Custom TCP' rule on port 3000, allowing traffic from '0.0.0.0/0'. The other rules are for SSH (port 22), MySQL/Aurora (port 3306), HTTPS (port 443), and HTTP (port 80).

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-007c80252138f7da	IPv4	SSH	TCP	22	0.0.0.0/0	-
☐	-	sgr-05c84101560c82b7b	IPv4	MySQL/Aurora	TCP	3306	0.0.0.0/0	-
☐	-	sgr-0fcb1acc8dd8f93de	IPv4	HTTPS	TCP	443	0.0.0.0/0	-
☐	-	sgr-0ba6db1a80da6b9ee	IPv4	HTTP	TCP	80	0.0.0.0/0	-
☐	-	sgr-0c145845f09015ed7	IPv4	Custom TCP	TCP	3000	0.0.0.0/0	-

Trouver le PID

ps aux | grep node

Arrêter l'application

kill